

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA1011	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	AKTCHAYA A	Reg. No.:	192424096
Date:	22/08/2026	Duration:	60 minutes

Code of Conduct

I AKTCHAYA A certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: AKTCHAYA A

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
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Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Test Case Design

University Academic Management and Student Success Platform

1. System Description

The University Academic Management and Student Success Platform is a centralized system designed to manage student admissions, academic records, course registration, attendance, assessments, examinations, faculty activities, notifications, academic reports, and student performance.

The system also identifies academically at-risk students based on configurable academic policies and provides notifications to students, faculty members, and academic advisors.

2. Functional Requirements

Requirement ID	Functional Requirement
FR01	The system shall manage student admission and registration.
FR02	The system shall maintain student academic records.
FR03	The system shall support course registration and withdrawal.
FR04	The system shall validate course prerequisites.
FR05	The system shall record and monitor student attendance.
FR06	The system shall manage assessments and examinations.
FR07	The system shall calculate marks, grades, and academic performance.
FR08	The system shall monitor student academic progress.
FR09	The system shall identify academically at-risk students.
FR10	The system shall notify students, faculty, and academic advisors.
FR11	The system shall generate academic and institutional reports.
FR12	The system shall provide dashboards for students, faculty, and administrators.
FR13	The system shall manage faculty activities.
FR14	The system shall provide secure authentication and role-based access.

3. Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR01	The system shall provide acceptable response time.
NFR02	The system shall support thousands of students and multiple departments.

NFR03	The system shall protect student and academic information.
NFR04	The system shall be reliable and prevent data loss.
NFR05	The system shall provide high availability.
NFR06	The system shall be easy to use and navigate.
NFR07	The system shall maintain data integrity and consistency.
NFR08	The system shall be compatible with supported browsers and devices.
NFR09	The system shall allow academic policies to be configured.
NFR10	The system shall support concurrent users without performance degradation.

4. Test Scenarios

Scenario ID	Test Scenario
TS01	Verify student admission
TS02	Verify student profile management
TS03	Verify academic record management
TS04	Verify course registration
TS05	Verify prerequisite validation
TS06	Verify duplicate course registration
TS07	Verify course capacity validation
TS08	Verify course withdrawal
TS09	Verify attendance recording
TS10	Verify attendance calculation
TS11	Verify attendance shortage detection
TS12	Verify assessment management
TS13	Verify marks entry
TS14	Verify grade calculation
TS15	Verify examination result management
TS16	Verify student performance monitoring
TS17	Verify at-risk student identification
TS18	Verify notification functionality
TS19	Verify academic report generation
TS20	Verify dashboard functionality

TS21	Verify faculty activity management
TS22	Verify authentication
TS23	Verify authorization
TS24	Verify system performance
TS25	Verify system scalability
TS26	Verify system reliability and recovery

5. Detailed Test Cases

- **Student Admission and Academic Records**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Register student with valid details	Login → Student Management → Add Student → Enter details → Submit	Valid Student ID, Name, DOB, Department	Student should be successfully registered	To be filled	Pass/Fail
TC02	Register student with missing name	Enter student details without name → Submit	Name = Blank	System should display mandatory field error	To be filled	Pass/Fail
TC03	Register student with invalid email	Enter invalid email → Submit	student@	System should reject invalid email	To be filled	Pass/Fail
TC04	Register duplicate student	Enter an existing student ID → Submit	Existing Student ID	Duplicate registration should be prevented	To be filled	Pass/Fail

TC05	Update student information	Search student → Edit → Update → Save	Updated phone/program	Updated information should be saved	To be filled	Pass/Fail
TC06	View academic record	Login → Student Profile → Academic Record	Valid Student ID	Correct academic information should be displayed	To be filled	Pass/Fail

- **Course Registration**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC07	Register eligible course	Login → Course Registration → Select course → Register	Eligible course	Course should be registered successfully	To be filled	Pass/Fail
TC08	Register course without prerequisite	Select course → Register	Prerequisite not completed	Registration should be rejected	To be filled	Pass/Fail
TC09	Register same course twice	Register course → Select same course → Register	Same course	Duplicate registration should be prevented	To be filled	Pass/Fail
TC10	Register full course	Select course with no available seats → Register	Capacity = 0	Registration should be rejected	To be filled	Pass/Fail

- **Attendance Management**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC11	Record attendance	Faculty Login → Course → Attendance → Save	Present	Attendance should be recorded	To be filled	Pass/Fail
TC12	Calculate attendance	Enter attendance → View percentage	8 Present, 2 Absent	Attendance should be 80%	To be filled	Pass/Fail
TC13	Detect attendance shortage	Record attendance below threshold	65%	Student should be identified as attendance deficient	To be filled	Pass/Fail
TC14	Enter invalid attendance	Enter invalid value	ABC	Invalid input should be rejected	To be filled	Pass/Fail
TC15	Test minimum attendance boundary	Enter attendance exactly at threshold	75%	System should apply configured attendance rule correctly	To be filled	Pass/Fail

- **Assessment and Examination**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC16	Create assessment	Faculty → Assessment → Create → Save	Valid assessment details	Assessment should be created	To be filled	Pass/Fail
TC17	Enter valid marks	Select student → Enter marks → Save	85/100	Marks should be saved	To be filled	Pass/Fail

TC18	Enter marks above maximum	Enter marks → Save	105/100	System should reject marks	To be filled	Pass/Fail
TC19	Enter negative marks	Enter marks → Save	-5	System should reject marks	To be filled	Pass/Fail
TC20	Calculate grade	Enter marks → Calculate	85	Correct grade should be generated	To be filled	Pass/Fail
TC21	Publish results	Select examination → Publish	Valid results	Results should be published to authorized users	To be filled	Pass/Fail

- Student Performance and At-Risk Identification**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC22	View student performance	Login → Performance Dashboard	Valid Student ID	Marks, grades, attendance and progress should be displayed	To be filled	Pass/Fail
TC23	Identify low-performing student	Enter performance below threshold	CGPA = 4.5	Student should be identified as at-risk	To be filled	Pass/Fail
TC24	Verify high-performing student	Enter performance above threshold	CGPA = 8.5	Student should not be marked at-risk	To be filled	Pass/Fail
TC25	Identify low-attendance student	Enter attendance	60%	Student should be identified as at-risk	To be filled	Pass/Fail

		below threshold				
TC26	Verify combined risk condition	Enter low attendance and low performance	CGPA = 4.5, Attendance = 60%	Student should be classified as at-risk and notification should be generated	To be filled	Pass/Fail

- **Notifications**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC27	Attendance notification	Trigger attendance shortage	Attendance = 60%	Notification should be sent to student and relevant faculty/advisor	To be filled	Pass/Fail
TC28	Examination notification	Create exam → Send notification	Exam details	Students should receive notification	To be filled	Pass/Fail
TC29	Verify notification content	Open notification	Valid notification	Correct message and details should be displayed	To be filled	Pass/Fail
TC30	Notification service failure	Disable notification service → Trigger alert	Service unavailable	System should handle failure without crashing	To be filled	Pass/Fail

- **Reports and Dashboards**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC31	Generate academic report	Select student → Generate Report	Valid student	Correct report should be generated	To be filled	Pass/Fail
TC32	Generate attendance report	Select course → Generate	Attendance data	Correct attendance report should be generated	To be filled	Pass/Fail
TC33	Generate department report	Admin → Reports → Department	Department data	Correct department report should be displayed	To be filled	Pass/Fail
TC34	Export report	Generate report → Export	PDF/CSV	Report should be exported successfully	To be filled	Pass/Fail

- **Authentication and Authorization**

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC36	Login with valid credentials	Enter username/password → Login	Valid credentials	User should successfully log in	To be filled	Pass/Fail
TC38	Login with incorrect password	Enter username and wrong password	Wrong password	Login should fail	To be filled	Pass/Fail
TC39	Login with blank fields	Leave fields blank → Login	Blank username/password	Validation message should be displayed	To be filled	Pass/Fail

TC40	Student accesses admin page	Login as student → Open admin page	Student account	Access should be denied	To be filled	Pass/Fail
TC41	Faculty accesses assigned course	Login as faculty → Open assigned course	Faculty account	Authorized course information should be accessible	To be filled	Pass/Fail
TC42	Logout	Login → Logout	Valid session	User should be logged out successfully	To be filled	Pass/Fail

7. Test Design Techniques

- **Equivalence Partitioning**

Example: Examination Marks

Valid range: 0–100

Partition	Test Data	Expected Result
Valid	50	Accept
Valid	0	Accept
Valid	100	Accept
Invalid	-1	Reject
Invalid	101	Reject
Invalid	ABC	Reject

- **Boundary Value Analysis**

Example: Attendance Requirement = 75%

Test Data	Condition	Expected Result
74%	Just below boundary	Attendance shortage
75%	Boundary value	Apply configured rule
76%	Just above boundary	Satisfy minimum requirement

- **Decision Table Testing**

At-Risk Student Identification

Condition	Rule 1	Rule 2	Rule 3	Rule 4
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Low Attendance	Yes	Yes	No	No
Low Academic Performance	Yes	No	Yes	No
Identify as At-Risk	Yes	Yes	Yes	No
Send Notification	Yes	Yes	Yes	No

- **State Transition Testing**

Course Registration

Available → Selected → Registered → Withdrawn

Current State	Action	Expected State
Available	Select Course	Selected
Selected	Confirm Registration	Registered
Registered	Withdraw	Withdrawn
Available	Register without prerequisite	Registration Rejected
Registered	Register same course again	Registration Rejected

8. Non-Functional Test Cases

TC ID	Test Scenario	Expected Result
NFT01	Test system with thousands of student records	System should remain stable
NFT02	Multiple users access system simultaneously	System should handle concurrent users
NFT03	Generate large institutional report	Report should be generated within acceptable time
NFT04	Attempt unauthorized access	Access should be denied
NFT05	Test system after network failure	System should recover without data loss
NFT06	Test supported browsers	System should function correctly
NFT07	Test navigation and usability	Users should easily navigate the system
NFT08	Update academic record	Data should remain accurate and consistent
NFT09	Test system during peak usage	System should remain available
NFT10	Test sensitive student information	Data should be protected from unauthorized access

9. Requirement Traceability Matrix

Requirement	Description	Covered Test Cases
FR01	Student Admission	TC01–TC06
FR02	Academic Records	TC05–TC06
FR03	Course Registration	TC07–TC11
FR04	Prerequisite Validation	TC08
FR05	Attendance	TC12–TC16
FR06	Assessment & Examination	TC17–TC22
FR07	Marks & Grades	TC18–TC21
FR08	Performance Monitoring	TC23–TC27
FR09	At-Risk Identification	TC24–TC27
FR10	Notifications	TC28–TC31
FR11	Reports	TC32–TC35
FR12	Dashboards	TC36
FR13	Faculty Activities	TC17–TC18
FR14	Authentication & Authorization	TC37–TC42
NFR01	Performance	NFT01–NFT03
NFR02	Scalability	NFT01–NFT02
NFR03	Security	NFT04, NFT10
NFR04	Reliability	NFT05
NFR05	Availability	NFT09
NFR06	Usability	NFT07
NFR07	Data Integrity	NFT08
NFR08	Compatibility	NFT06

10. Test Coverage Summary

Testing Area	Coverage
Student Management	Covered
Academic Records	Covered
Course Registration	Covered
Attendance Management	Covered
Assessments	Covered
Examinations	Covered

Student Performance	Covered
At-Risk Identification	Covered
Notifications	Covered
Reports	Covered
Dashboards	Covered
Faculty Activities	Covered
Authentication	Covered
Authorization	Covered
Security	Covered
Performance	Covered
Scalability	Covered
Reliability	Covered
Usability	Covered
Compatibility	Covered

11. Expected Outcome

The designed test cases verify that the University Academic Management and Student Success Platform:

1. Correctly manages student academic records.
2. Supports secure student admission and course registration.
3. Accurately records and calculates attendance.
4. Correctly manages assessments and examinations.
5. Accurately calculates marks and grades.
6. Monitors student academic performance.
7. Correctly identifies academically at-risk students.
8. Provides timely notifications to relevant stakeholders.
9. Generates accurate academic and institutional reports.
10. Provides appropriate dashboards for different users.
11. Prevents unauthorized access to academic information.
12. Handles invalid, boundary, and exceptional inputs correctly.
13. Supports thousands of students and multiple departments.
14. Maintains reliability, security, data integrity, and availability.

15. Provides a usable and efficient platform for students, faculty, advisors, and administrators.

12. Conclusion

The test case design provides comprehensive functional and non-functional coverage for the **University Academic Management and Student Success Platform**. The test cases cover normal, invalid, boundary, and exceptional conditions. **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing** are applied to improve test effectiveness and ensure that the proposed system meets its functional, security, reliability, scalability, usability, and performance requirements.